TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	72.6 ± 2.91
Correlation Stability	92.5 ± 1.28
Distribution Consistency	85.9 ± 1.04
Entropy Preservation	72.0 ± 0.00
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	96.1 ± 1.76
Privacy Risk	100.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	0.0 ± 0.00
Feature Accuracy	95.3 ± 1.10
Novelty Score	0.0 ± 0.00

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Dataset Audit: breast_cancer_wisconsin_diagnostic.csv

1. Dataset Portrait

Attribute	Value
Rows	569
Columns	31
Numeric features	30
Categorical features	1
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	72.6 ± 2.91
Correlation Stability (*2)	92.5 ± 1.28
Distribution Consistency (*3)	85.9 ± 1.04
Entropy Preservation (*4)	72.0 ± 0.00
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	96.1 ± 1.76
Privacy Risk (*7)	100.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	0.0 ± 0.00
Feature Accuracy (*10)	95.3 ± 1.10
Novelty Score (*11)	0.0 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9342105263157896	N/A	0.9368421052631579	0.0026315789473684007	
F1	0.9250062311022852	N/A	0.926735280483172	0.0017290493808868822	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9605263157894737	N/A	0.9614035087719298	0.0008771929824561431	
F1	0.9556894690567029	N/A	0.9564737827821931	0.0007843137254901933	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9394736842105263	N/A	0.9394736842105263	0	
F1	0.9314283604116784	N/A	0.9314283604116784	0	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9333333333333333	N/A	0.9333333333333333	0	
F1	0.9242527459470589	N/A	0.9242527459470589	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9692982456140351	N/A	0.9692982456140351	0	
F1	0.9657812227588884	N/A	0.9657812227588884	0	

Model: KNN

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9429824561403507	N/A	0.9429824561403507	0	
F1	0.934337828992506	N/A	0.934337828992506	0	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9508771929824562	N/A	0.9508771929824562	0	
F1	0.9452155271956949	N/A	0.9452155271956949	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9456140350877194	N/A	0.9324561403508772	-0.013157894736842113	
F1	0.9369599527876111	N/A	0.9248157326366462	-0.012144220150965035	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9710526315789474	N/A	0.9710526315789474	0	
F1	0.9676338620041433	N/A	0.9676338620041433	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9561403508771928	N/A	0.9605263157894737	0.004385964912280715	
F1	0.9500474168878243	N/A	0.9548037304519953	0.004756313564170889	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	17.99	10.38	122.8	1001.0	0.1184	0.2776	0.3001	0.1471	0.2419	0.07871
2	20.57	17.77	132.9	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667
3	19.69	21.25	130.0	1203.0	0.1096	0.1599	0.1974	0.1279	0.2069	0.05999
4	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0.1052	0.2597	0.09744
5	20.29	14.34	135.1	1297.0	0.1003	0.1328	0.198	0.1043	0.1809	0.05883
6	12.45	15.7	82.57	477.1	0.1278	0.17	0.1578	0.08089	0.2087	0.07613
7	18.25	19.98	119.6	1040.0	0.09463	0.109	0.1127	0.074	0.1794	0.05742
8	13.71	20.83	90.2	577.9	0.1189	0.1645	0.09366	0.05985	0.2196	0.07451
9	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
10	12.46	24.04	83.97	475.9	0.1186	0.2396	0.2273	0.08543	0.203	0.08243
11	16.02	23.24	102.7	797.8	0.08206	0.06669	0.03299	0.03323	0.1528	0.05697
12	15.78	17.89	103.6	781.0	0.0971	0.1292	0.09954	0.06606	0.1842	0.06082
13	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
14	15.85	23.95	103.7	782.7	0.08401	0.1002	0.09938	0.05364	0.1847	0.05338
15	13.73	22.61	93.6	578.3	0.1131	0.2293	0.2128	0.08025	0.2069	0.07682
16	14.54	27.54	96.73	658.8	0.1139	0.1595	0.1639	0.07364	0.2303	0.07077
17	14.68	20.13	94.74	684.5	0.09867	0.072	0.07395	0.05259	0.1586	0.05922
18	16.13	20.68	108.1	798.8	0.117	0.2022	0.1722	0.1028	0.2164	0.07356
19	19.81	22.15	130.0	1260.0	0.09831	0.1027	0.1479	0.09498	0.1582	0.05395
20	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	0.04781	0.1885	0.05766

B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	14.27	22.55	93.77	629.8	0.1038	0.1154	0.1463	0.06139	0.1926	0.05982
2	16.13	17.88	107.0	807.2	0.104	0.1559	0.1354	0.07752	0.1998	0.06515
3	11.29	13.04	72.23	388.0	0.09834	0.07608	0.03265	0.02755	0.1769	0.0627
4	11.34	21.26	72.48	396.5	0.08759	0.06575	0.05133	0.01899	0.1487	0.06529
5	12.86	13.32	82.82	504.8	0.1134	0.08834	0.038	0.034	0.1543	0.06476
6	9.333	21.94	59.01	264.0	0.0924	0.05605	0.03996	0.01282	0.1692	0.06576
7	13.24	20.13	86.87	542.9	0.08284	0.1223	0.101	0.02833	0.1601	0.06432
8	11.2	29.37	70.67	386.0	0.07449	0.03558	0.0	0.0	0.106	0.05502
9	16.74	21.59	110.1	869.5	0.0961	0.1336	0.1348	0.06018	0.1896	0.05656
10	13.17	18.66	85.98	534.6	0.1158	0.1231	0.1226	0.0734	0.2128	0.06777

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	19.59	25.0	127.7	1191.0	0.1032	0.09871	0.1655	0.09063	0.1663	0.05391
2	11.49	14.59	73.99	404.9	0.1046	0.08228	0.05308	0.01969	0.1779	0.06574
3	10.6	18.95	69.28	346.4	0.09688	0.1147	0.06387	0.02642	0.1922	0.06491
4	19.0	18.91	123.4	1138.0	0.08217	0.08028	0.09271	0.05627	0.1946	0.05044
5	12.27	29.97	77.42	465.4	0.07699	0.03398	0.0	0.0	0.1701	0.0596
6	17.02	23.98	112.8	899.3	0.1197	0.1496	0.2417	0.1203	0.2248	0.06382
7	13.28	20.28	87.32	545.2	0.1041	0.1436	0.09847	0.06158	0.1974	0.06782
8	20.59	21.24	137.8	1320.0	0.1085	0.1644	0.2188	0.1121	0.1848	0.06222
9	12.18	20.52	77.22	458.7	0.08013	0.04038	0.02383	0.0177	0.1739	0.05677
10	19.73	19.82	130.7	1206.0	0.1062	0.1849	0.2417	0.0974	0.1733	0.06697

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	20.6	29.33	140.1	1265.0	0.1178	0.277	0.3514	0.152	0.2397	0.07016
2	7.691	25.44	48.34	170.4	0.08668	0.1199	0.09252	0.01364	0.2037	0.07751
3	11.26	19.96	73.72	394.1	0.0802	0.1181	0.09274	0.05588	0.2595	0.06233
4	16.25	19.51	109.8	815.8	0.1026	0.1893	0.2236	0.09194	0.2151	0.06578
5	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
6	20.44	21.78	133.8	1293.0	0.0915	0.1131	0.09799	0.07785	0.1618	0.05557
7	10.97	17.2	71.73	371.5	0.08915	0.1113	0.09457	0.03613	0.1489	0.0664
8	17.29	22.13	114.4	947.8	0.08999	0.1273	0.09697	0.07507	0.2108	0.05464
9	20.31	27.06	132.9	1288.0	0.1	0.1088	0.1519	0.09333	0.1814	0.05572
10	11.9	14.65	78.11	432.8	0.1152	0.1296	0.0371	0.03003	0.1995	0.07839

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.81	23.75	91.56	597.8	0.1323	0.1768	0.1558	0.09176	0.2251	0.07421
2	13.01	22.22	82.01	526.4	0.06251	0.01938	0.001595	0.001852	0.1395	0.05234
3	17.99	20.66	117.8	991.7	0.1036	0.1304	0.1201	0.08824	0.1992	0.06069
4	11.64	18.33	75.17	412.5	0.1142	0.1017	0.0707	0.03485	0.1801	0.0652
5	13.85	19.6	88.68	592.6	0.08684	0.0633	0.01342	0.02293	0.1555	0.05673
6	12.88	18.22	84.45	493.1	0.1218	0.1661	0.04825	0.05303	0.1709	0.07253
7	20.09	23.86	134.7	1247.0	0.108	0.1838	0.2283	0.128	0.2249	0.07469
8	11.52	18.75	73.34	409.0	0.09524	0.05473	0.03036	0.02278	0.192	0.05907
9	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
10	17.75	28.03	117.3	981.6	0.09997	0.1314	0.1698	0.08293	0.1713	0.05916

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	11.6	18.36	73.88	412.7	0.08508	0.05855	0.03367	0.01777	0.1516	0.05859
2	11.93	10.91	76.14	442.7	0.08872	0.05242	0.02606	0.01796	0.1601	0.05541
3	20.16	19.66	131.1	1274.0	0.0802	0.08564	0.1155	0.07726	0.1928	0.05096
4	13.0	25.13	82.61	520.2	0.08369	0.05073	0.01206	0.01762	0.1667	0.05449
5	10.2	17.48	65.05	321.2	0.08054	0.05907	0.05774	0.01071	0.1964	0.06315
6	20.26	23.03	132.4	1264.0	0.09078	0.1313	0.1465	0.08683	0.2095	0.05649
7	8.726	15.83	55.84	230.9	0.115	0.08201	0.04132	0.01924	0.1649	0.07633
8	19.4	18.18	127.2	1145.0	0.1037	0.1442	0.1626	0.09464	0.1893	0.05892
9	19.73	19.82	130.7	1206.0	0.1062	0.1849	0.2417	0.0974	0.1733	0.06697
10	12.19	13.29	79.08	455.8	0.1066	0.09509	0.02855	0.02882	0.188	0.06471

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	12.05	22.72	78.75	447.8	0.06935	0.1073	0.07943	0.02978	0.1203	0.06659
2	12.87	16.21	82.38	512.2	0.09425	0.06219	0.039	0.01615	0.201	0.05769
3	9.904	18.06	64.6	302.4	0.09699	0.1294	0.1307	0.03716	0.1669	0.08116
4	16.78	18.8	109.3	886.3	0.08865	0.09182	0.08422	0.06576	0.1893	0.05534
5	23.21	26.97	153.5	1670.0	0.09509	0.1682	0.195	0.1237	0.1909	0.06309
6	18.08	21.84	117.4	1024.0	0.07371	0.08642	0.1103	0.05778	0.177	0.0534
7	11.6	12.84	74.34	412.6	0.08983	0.07525	0.04196	0.0335	0.162	0.06582
8	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
9	15.46	11.89	102.5	736.9	0.1257	0.1555	0.2032	0.1097	0.1966	0.07069
10	14.9	22.53	102.1	685.0	0.09947	0.2225	0.2733	0.09711	0.2041	0.06898

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.5	12.71	85.69	566.2	0.07376	0.03614	0.002758	0.004419	0.1365	0.05335
2	23.27	22.04	152.1	1686.0	0.08439	0.1145	0.1324	0.09702	0.1801	0.05553
3	14.9	22.53	102.1	685.0	0.09947	0.2225	0.2733	0.09711	0.2041	0.06898
4	13.16	20.54	84.06	538.7	0.07335	0.05275	0.018	0.01256	0.1713	0.05888
5	14.26	18.17	91.22	633.1	0.06576	0.0522	0.02475	0.01374	0.1635	0.05586
6	8.598	20.98	54.66	221.8	0.1243	0.08963	0.03	0.009259	0.1828	0.06757
7	19.89	20.26	130.5	1214.0	0.1037	0.131	0.1411	0.09431	0.1802	0.06188
8	17.42	25.56	114.5	948.0	0.1006	0.1146	0.1682	0.06597	0.1308	0.05866
9	11.95	14.96	77.23	426.7	0.1158	0.1206	0.01171	0.01787	0.2459	0.06581
10	14.69	13.98	98.22	656.1	0.1031	0.1836	0.145	0.063	0.2086	0.07406

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	11.47	16.03	73.02	402.7	0.09076	0.05886	0.02587	0.02322	0.1634	0.06372
2	10.57	18.32	66.82	340.9	0.08142	0.04462	0.01993	0.01111	0.2372	0.05768
3	13.46	18.75	87.44	551.1	0.1075	0.1138	0.04201	0.03152	0.1723	0.06317
4	12.7	12.17	80.88	495.0	0.08785	0.05794	0.0236	0.02402	0.1583	0.06275
5	14.64	16.85	94.21	666.0	0.08641	0.06698	0.05192	0.02791	0.1409	0.05355
6	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0.1052	0.2597	0.09744
7	13.05	18.59	85.09	512.0	0.1082	0.1304	0.09603	0.05603	0.2035	0.06501
8	10.16	19.59	64.73	311.7	0.1003	0.07504	0.005025	0.01116	0.1791	0.06331
9	16.24	18.77	108.8	805.1	0.1066	0.1802	0.1948	0.09052	0.1876	0.06684
10	14.87	20.21	96.12	680.9	0.09587	0.08345	0.06824	0.04951	0.1487	0.05748

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	11.5	18.45	73.28	407.4	0.09345	0.05991	0.02638	0.02069	0.1834	0.05934
2	12.65	18.17	82.69	485.6	0.1076	0.1334	0.08017	0.05074	0.1641	0.06854
3	14.86	23.21	100.4	671.4	0.1044	0.198	0.1697	0.08878	0.1737	0.06672
4	15.1	22.02	97.26	712.8	0.09056	0.07081	0.05253	0.03334	0.1616	0.05684
5	15.71	13.93	102.0	761.7	0.09462	0.09462	0.07135	0.05933	0.1816	0.05723
6	15.08	25.74	98.0	716.6	0.1024	0.09769	0.1235	0.06553	0.1647	0.06464
7	14.97	16.95	96.22	685.9	0.09855	0.07885	0.02602	0.03781	0.178	0.0565
8	11.43	15.39	73.06	399.8	0.09639	0.06889	0.03503	0.02875	0.1734	0.05865
9	11.81	17.39	75.27	428.9	0.1007	0.05562	0.02353	0.01553	0.1718	0.0578
10	10.57	18.32	66.82	340.9	0.08142	0.04462	0.01993	0.01111	0.2372	0.05768

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	15.12	16.68	98.78	716.6	0.08876	0.09588	0.0755	0.04079	0.1594	0.05986
2	15.37	22.76	100.2	728.2	0.092	0.1036	0.1122	0.07483	0.1717	0.06097
3	20.18	19.54	133.8	1250.0	0.1133	0.1489	0.2133	0.1259	0.1724	0.06053
4	19.53	32.47	128.0	1223.0	0.0842	0.113	0.1145	0.06637	0.1428	0.05313
5	13.81	23.75	91.56	597.8	0.1323	0.1768	0.1558	0.09176	0.2251	0.07421
6	20.57	17.77	132.9	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667
7	13.71	18.68	88.73	571.0	0.09916	0.107	0.05385	0.03783	0.1714	0.06843
8	11.94	18.24	75.71	437.6	0.08261	0.04751	0.01972	0.01349	0.1868	0.0611
9	10.51	23.09	66.85	334.2	0.1015	0.06797	0.02495	0.01875	0.1695	0.06556
10	11.75	17.56	75.89	422.9	0.1073	0.09713	0.05282	0.0444	0.1598	0.06677